DAY 6 | HANDS-ON ACTIVITIES

SQLAlchemy, Relationships and API Docs: Problem Statements

Seven problems that grow one Library API. SQLite works the same in Colab and VS Code.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Spot the Relationship

Level: Warm-up | **Concepts:** relationship design

For each pair, write the relationship type and where the foreign key or link table goes: customer and ticket, ticket and comment, user and profile, agent and ticket, book and tag.

Sample run / expected result

```
customer - ticket: one-to-many, FK customer_id on tickets
book - tag: many-to-many, link table book_tags
```

Hints

- Ask how many of each side the other can have.

Problem 2: First ORM Model

Level: Easy | **Concepts:** engine, model, session, select

Create a BookRow model in SQLite. Insert 5 books, query those with more than 200 pages sorted by title, update one, and delete one.

Sample run / expected result

```
Books over 200 pages: Atomic Habits, Deep Work, Ikigai
```

Hints

- select(BookRow).where(BookRow.pages > 200).order_by(BookRow.title)

Problem 3: Authors and Books

Level: Easy | **Concepts:** one-to-many, back_populates

Create Author and Book models with a one-to-many relationship. Add 3 authors with 2 books each, then print each author with their book titles, and all books by one author using a join.

Sample run / expected result

```
Ito: Tokyo Nights, Rain in Kyoto
Rao: Monsoon Code, Chai Stories
```

Hints

- relationship(back_populates=...) on both classes

Problem 4: Constraints in Action

Level: Medium | **Concepts:** unique, nullable, IntegrityError

Make author name unique and book title not nullable. Try to break each rule and catch IntegrityError, rolling back each time.

Sample run / expected result

```
Duplicate author -> IntegrityError, rolled back
Missing title -> IntegrityError, rolled back
```

Hints

- from sqlalchemy.exc import IntegrityError

Problem 5: Safe Transfer

Level: Challenging | **Concepts:** transactions, rollback

Create an Account table. Write transfer(from_id, to_id, amount) that rolls back if the balance is too low or an account is missing. Prove the balances are unchanged after a failed transfer.

Sample run / expected result

```
transfer(1, 2, 9999) -> failed, balances still 1000 and 500
```

Hints

- Do both updates, then commit once at the end.

Problem 6: Nested Responses

Level: Challenging | **Concepts:** FastAPI + relationships + Pydantic

Add GET /authors/{id} returning the author with a list of their books, and GET /books returning each book with its author name. Return 404 for a missing author.

Sample run / expected result

```
{"id": 1, "name": "Ito", "books": [{"id": 1, "title": "Tokyo Nights"}]}
```

Hints

- Nested Pydantic models with from_attributes=True
- selectinload to avoid N+1

Problem 7: Docs a Teammate Can Use

Level: Stretch | **Concepts:** OpenAPI metadata, Codex

Add app title, version and description; tags with descriptions; summary and 404 responses on every endpoint; and examples on models. You may ask Codex to draft them; check each one matches the code. Export /openapi.json to a file.

Sample run / expected result

```
openapi.json saved, /docs shows grouped, described endpoints
```

Hints

- FastAPI(openapi_tags=[...])
- json.dump(app.openapi(), f, indent=2)